INDONESIA · CARDIOMETABOLIC DISEASE · FLAGSHIP REPORT 1 OF 3



Traditional dishes at a food stall in West Java. Photo: Faisal (@Photobapak), Pexels

Risk Factors and Prevention Policy

Are prevention policies keeping pace with rising risks?

23.0%
of adults with obesity (BMI 27+), 2023

317
of 464 comparable districts where obesity rose significantly, 2013-2023

43.8%
of men aged 10 and over smoke every day, 2023

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FROM THE REPORT

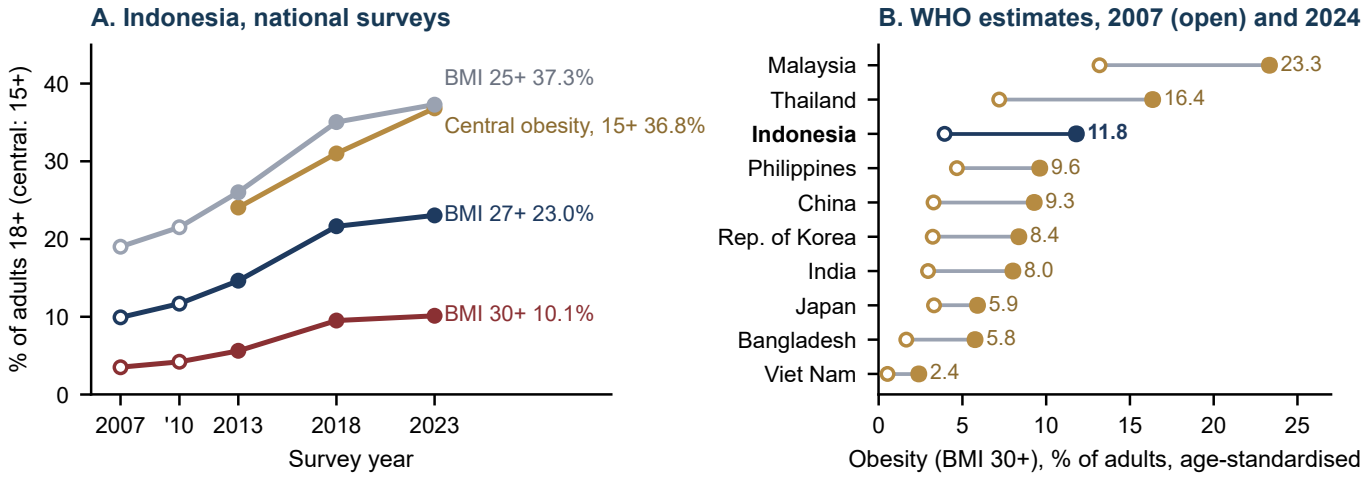
Chapter 3, pages 36-51. AMHASS Flagship Report NCD-FL-01, Report 1 of 3.

Risk Factor 1: Overweight and Obesity

Obesity has more than doubled since 2007.

EXHIBIT 24

Adult obesity (BMI 27+) rose from 9.9% (2007) to 23.0% (2023), 2.1-fold since 2013 in the poorest



A: survey estimates with 95% CIs (confidence intervals, the likely range). Hollow dots (2007, 2010) are close to, but not identical in, method. The 2007 waist measurement is not comparable and is not drawn. B: WHO modelled estimates, using another method.

Table 10. Obesity (BMI 27 or more) among adults aged 18 and over, by group, %

Group	2013	2018	2023	Change 2013–2023, points
Women	19.7	29.0	30.7	+10.9
Men	9.5	14.3	15.4	+5.9
Urban	17.8	24.8	25.9	+8.1
Rural	11.4	17.7	19.0	+7.6
Poorest quintile	7.0	15.1	14.8	+7.8
Richest quintile	21.2	28.1	30.1	+8.9

Wealth quintiles: five equal groups of a household asset index built by principal component analysis (ARC harmonisation of Riskesdas and SKI). Design-based estimates.

In 2023, 23.0% of adults aged 18 and over had a body mass index (BMI) of 27 or more, up from 9.9% in 2007. Adult obesity has increased steadily in Indonesia, although growth has slowed since 2018. Using the Ministry of Health's definition of obesity (body mass index [BMI] ≥27), prevalence rose from 9.9% in 2007 to 14.6% in 2013, 21.6% in 2018, and 23.0% in 2023 (Riskesdas 2007–2018; SKI 2023). The increase slowed from 7.0 percentage points in 2013–2018 to 1.4 points in 2018–2023. However, the broader burden of excess weight remains substantial: central obesity, defined by excess fat around the waist, reached 36.8% of people aged 15 and older in 2023, up from 24.1% in 2013.

The burden is increasingly affecting women, poorer households, and rural communities. In 2023, obesity prevalence reached 30.7% among women, compared with 15.4% among men. The gender gap widened from 10.2 percentage points in 2013 to 15.2 points in 2023. Obesity remains more prevalent among wealthier adults, rising from 14.8% in the poorest quintile to 30.1% in the richest. Yet growth has been substantial across the socioeconomic distribution: prevalence in the poorest quintile increased 2.1-fold since 2013, compared with 1.4-fold in the richest.

Rural prevalence also grew faster than urban prevalence, increasing 1.67-fold versus 1.46-fold, consistent with patterns identified in earlier longitudinal research[1]. International comparisons reinforce the urgency of the trend. WHO modelled, age-standardised estimates using the international BMI threshold of 30 or higher show that adult obesity prevalence increased from 3.9% in 2007 to 11.8% in 2024. This 7.9-percentage-point increase was the third largest among the ten Asian countries compared, behind Malaysia and Thailand. Indonesia's free health check programme (*Cek Kesehatan Gratis*, CKG) provides an opportunity to identify obesity through routine measurement of weight and height among adults reached by the programme[2]. However, screening alone will not reverse the trend. Districts also need policy measures addressing the wider determinants of obesity, including those outlined in Exhibit 37, with particular attention to poorer and rural populations.

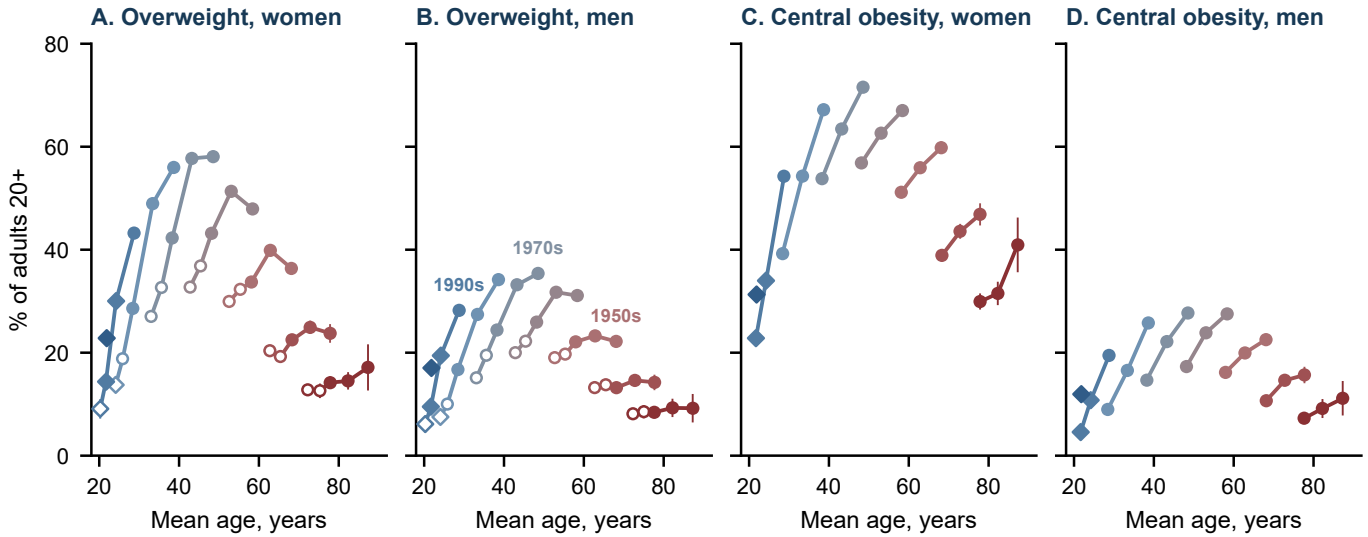
Sources: Ministry of Health, Riskesdas and SKI, harmonised by ARC; WHO Global Health Observatory, 2026 publication.

Risk Factor 1: Overweight and Waist by Generation

At about 38, 56.0% of women born in the 1980s were overweight, against 42.3% of women born in the 1970s.

EXHIBIT 25

At the same age, each generation of adults is more often overweight and wider at the waist



Lines: ten-year birth generations at their mean age in each survey (2007 to 2023), joining surveys, not people; labels: decade of birth; dark red: born earliest, through grey, to navy: born latest. Hollow: 2007 and 2010, weights only; diamonds: generations only partly aged 20 or more. Whiskers: 95% CIs. Overweight: BMI 25 or more; central obesity: waist 80 cm or more (women), 90 cm or more (men); pregnant women excluded where recorded[3].

Table 11. Overweight and central obesity at the same age, a generation apart, % of adults

Measure	At 28: born 1980s, 2013	At 28: born 1990s, 2023	At 38: born 1970s, 2013	At 38: born 1980s, 2023	At 48: born 1960s, 2013	At 48: born 1970s, 2023
Overweight, women	28.6	43.2	42.3	56.0	43.2	58.1
Overweight, men	16.7	28.2	24.4	34.2	25.9	35.4
Central obesity, women	39.2	54.3	53.8	67.2	56.9	71.6
Central obesity, men	9.0	19.5	14.7	25.8	17.3	27.7

Mean ages 28.4 to 28.8, 38.2 to 38.7 and 48.1 to 48.5: two generations ten years apart, measured at the same age in 2013 and 2023, the two surveys with the same design and height reading. Design-based estimates.

Following people born in the same decade through five national surveys shows each generation gaining weight through adult life and each reaching a given age heavier than the one before[3]. At about 38, 56.0% of women born in the 1980s were overweight (a BMI of 25 or more) in 2023, against 42.3% of women born in the 1970s at that age in 2013; at about 28, 43.2% against 28.6%. Men follow the same order at lower levels: 34.2% against 24.4% at about 38.

The waist shows the same order. Central obesity (a waist of 80 cm or more in women, 90 cm or more in men) reached 67.2% of women born in the 1980s at about 38, against 53.8% of women born in the 1970s at that age, and 25.8% of men against 14.7%. At the same BMI, the mean waist was 1.0 to 3.2 cm larger in 2023 than in 2013 at every adult age, a pattern that a change in how the waist was measured would also produce.

Overweight here starts at a BMI of 25 and at age 20, a wider measure than the obesity of Exhibit 24 (BMI 27 or more, 18+); the central obesity in the same exhibit (36.8% in 2023) counts people 15 and over with a waist above 80 or 90 cm. The surveys sample different people, so no person is followed; deaths and migration are not adjusted, and age, survey year and generation are not separated. The 2018 survey read height about 0.7 cm low, which raises its BMI by about 0.2; the 2007 waist is not comparable and is not drawn. **Each generation becomes overweight younger, so prevention has to start in young adulthood**, years before the rise in blood pressure by generation (Exhibit 40).

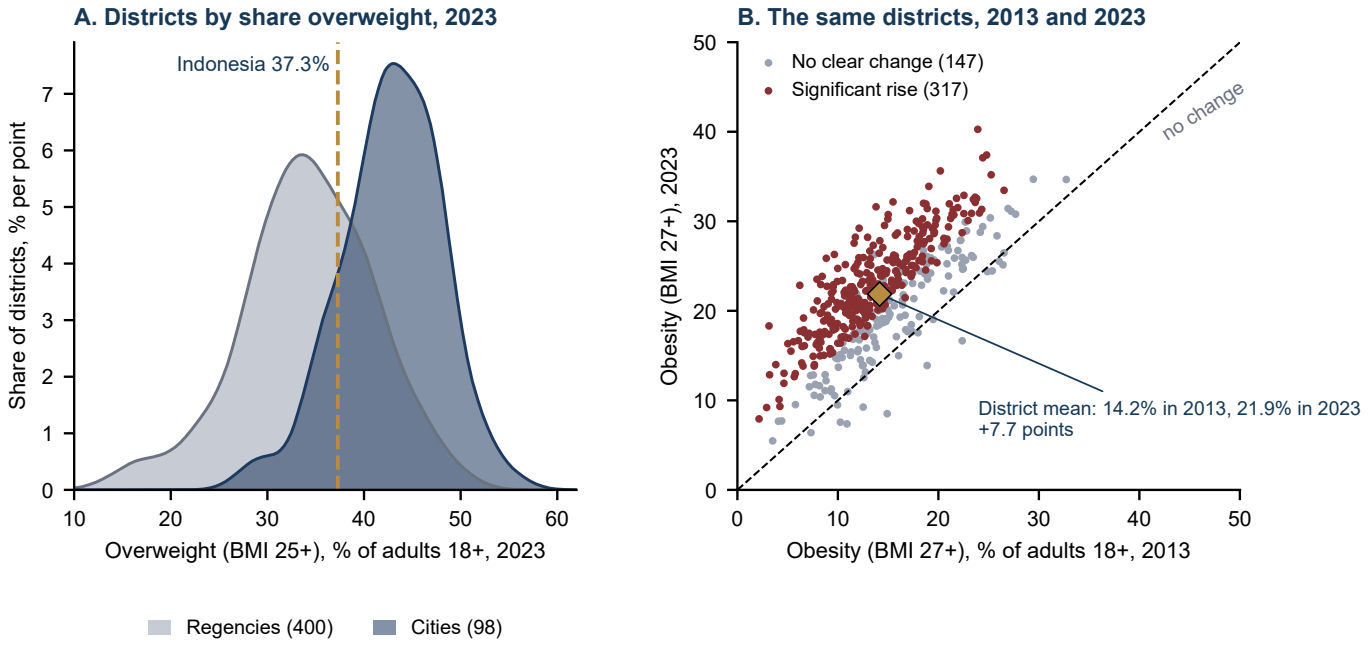
Sources: Ministry of Health, Riskesdas 2007, 2010, 2013 and 2018 and SKI 2023; ARC analysis of nutrition across the life course.

Risk Factor 1: Overweight and Obesity by District

Overweight is common and obesity is spreading across districts.

EXHIBIT 26

Adult obesity rose significantly in 317 of 464 districts between 2013 and 2023 and fell in none



A: AMHASS district estimates, SKI 2023, smoothed (kernel density: a smooth curve drawn through the data). B: each dot is a district with unchanged boundaries; dark red dot: a clear rise (95% CIs do not overlap); dashed: no change.

Table 12. The overweight ladder among adults aged 18 and over, 2023

Step	Indonesia, %	95% CI	Kept from step before, %	Median district	Median city	Median regency	Districts, 10th-90th percentile
Overweight (BMI 25+)	37.3	37.0 to 37.5		35.9	43.0	34.2	27.1 to 45.6
+ central obesity	27.5	27.3 to 27.7	73.7	25.7	33.5	24.5	17.4 to 35.0
+ high blood pressure	12.3	12.2 to 12.5	44.9	10.5	13.3	10.0	6.3 to 15.7

Each step includes the one before. Kept from step before: a ratio of two estimates, without interval. High blood pressure: 140/90 mmHg or above, measured. Districts with a published estimate: 498.

Among Indonesian adults who were overweight in 2023, 73.7% also had central obesity, and 44.9% of those with central obesity had raised blood pressure. Overall, 37.3% of adults aged 18 and older were overweight; 27.5% had both overweight and central obesity, while 12.3% had all three risks, including blood pressure of at least 140/90 mmHg (Ministry of Health, SKI 2023; AMHASS, ARC's district assessment of the survey).

Although overweight is more prevalent in cities, the need for action extends across the country. District-level prevalence ranged from 27.1% in the lowest tenth of districts to 45.6% in the highest. The median city had a prevalence of 43.0%, compared with 34.2% in the median regency. Provincial estimates ranged from 23.5% in Nusa Tenggara Timur to 47.4% in DKI Jakarta.

Yet 56.0% of the variation occurred between districts within the same province, meaning provincial averages can conceal substantial local differences. The increase has also been geographically widespread. Among the 464 districts with unchanged boundaries, obesity prevalence rose significantly in 317 (68.3%) between 2013 and 2023. The median district increased by 7.8 percentage points, and 122 districts recorded increases of at least 10 points. Central obesity also increased in 348 districts. The free health check programme (*Cek Kesehatan Gratis*, CKG) can provide an entry point, but weight and waist measurements should be linked to tailored advice, risk assessment, and follow-up in both regencies and cities.

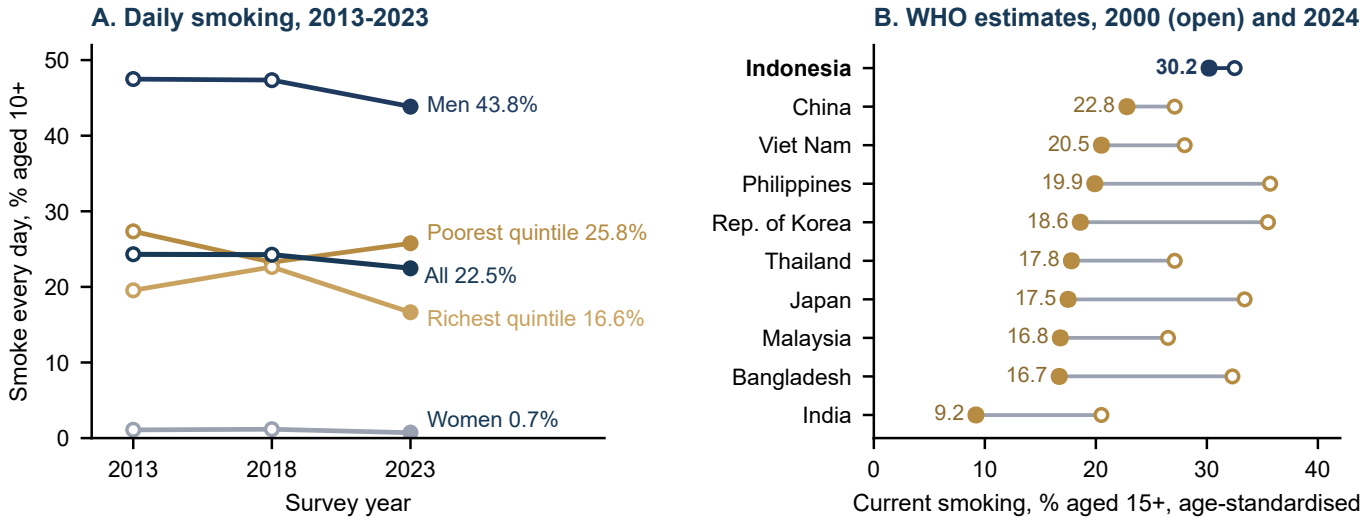
Sources: Ministry of Health, SKI 2023, AMHASS district estimates; Riskesdas 2013 and SKI 2023, harmonised district estimates, ARC analysis.

Risk Factor 2: Smoking

Smoking is falling slowly but remains high.

EXHIBIT 27

Daily smoking at ages 10+ fell after 2018 in every group but the poorest quintile, where it rose



A: survey estimates; hollow dots: method close to, but not identical with, 2023; 2018 to 2023, all and the two quintiles: 95% CIs do not overlap. Wealth quintiles: five equal groups of households, ranked by an index of household assets (built with principal component analysis). B: WHO modelled estimates, adjusted for age (age-standardised), for people aged 15 and over; a different population and method.

Table 13. Daily smoking among people aged 10 and over, by group, %

Group	2013	2018	2023	Change 2013–2023, points
Men	47.5	47.3	43.8	−3.7
Women	1.1	1.2	0.7	−0.4
Poorest quintile	27.3	23.3	25.8	−1.6
Richest quintile	19.5	22.6	16.6	−2.9
Aged 15–24	19.4	20.0	17.8	−1.6
Aged 65+	21.7	20.1	17.3	−4.4

2013 and 2018 are close to 2023 in method; 2007 asked about smoking or chewing and is not compared. Design-based estimates.

Daily smoking fell nationally after 2018, but not among the poorest households. Between 2018 and 2023, daily smoking rose from 23.3% to 25.8% in the poorest quintile, while falling from 22.6% to 16.6% in the richest. The gap widened to 9.1 percentage points. Overall, 22.5% of people aged 10 and over smoked daily in 2023, down from 24.3% in both 2013 and 2018 (Ministry of Health, Riskesdas and SKI).

Indonesia recorded the smallest decline in smoking among ten Asian countries and remains the highest. WHO age-standardised estimates put current smoking among people aged 15 and over at 30.2% in 2024, compared with 22.8% in China. Since 2000, Indonesia's prevalence fell by just 2.3 percentage points. Within Indonesia, the median district had 21.4% daily smokers, with higher prevalence in regencies (21.9%) than cities (18.5%). Of 464 districts with unchanged boundaries, current smoking fell significantly in 112 and rose in four between 2013 and 2023 (AMHASS district estimates).

Policy needs to address both affordability and access to cessation support. Tobacco prices should rise faster than incomes, while districts expand quitting support, particularly for poorer households and regencies. The national development plan (RPJMN) targets reducing smoking among people aged 10–21 from 12.4% in 2023 to 8.4% by 2029[4]. Smoking-related illness cost BPJS Kesehatan an estimated Rp10.4–15.6 trillion in 2019[5].

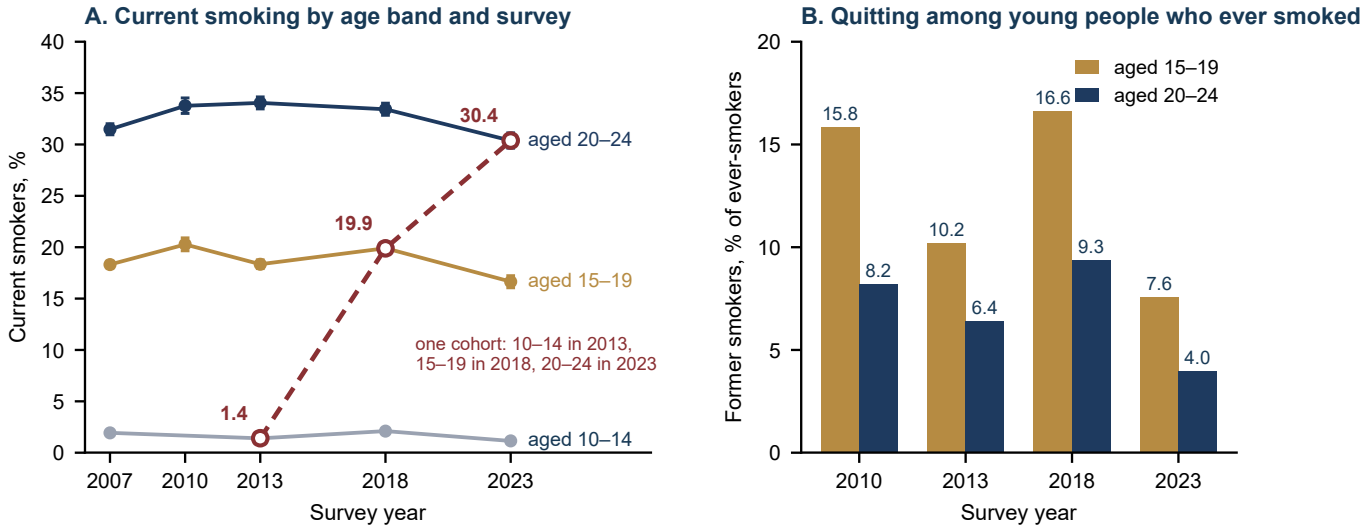
Sources: Ministry of Health, Riskesdas and SKI (ARC harmonisation), AMHASS district estimates; WHO; RPJMN 2025–2029.

Risk Factor 2: Smoking Uptake by Generation

One generation went from 1.4% smoking at ages 10 to 14 in 2013 to 30.4% at ages 20 to 24 in 2023.

EXHIBIT 28

Smoking is taken up between ages 10 and 24, and fewer young smokers quit in 2023 than in 2018



Current smokers: daily or occasional, a wider measure than the daily smoking of Exhibit 27. A: survey estimates with 95% CIs; dark red: the same birth years seen in three independent surveys, at 10 to 14 in 2013, 15 to 19 in 2018 and 20 to 24 in 2023 (a synthetic cohort, not the same people; deaths and migration not adjusted); Riskesdas 2010 asked only from age 15. B: former smokers as a share of everyone in the age band who had ever smoked; 2007 is left out because its split of former and never smokers does not match the published table.

Table 14. Smoking status of people aged 20 to 24, %

Status	2010	2013	2018	2023
Daily smoker	26.1	27.2	27.5	24.5
Occasional smoker	7.7	6.8	5.9	5.9
Former smoker	3.0	2.3	3.5	1.3
Never smoked	63.2	63.6	63.1	68.3
Former smokers, % of ever-smokers	8.2	6.4	9.3	4.0

Daily and occasional smokers together are the current smokers of panel A. Design-based estimates. 2007 is not shown: its split of former and never smokers does not match the published table.

Smoking starts in adolescence and climbs through the early twenties. The birth years aged 10 to 14 in 2013 smoked at 1.4%; the same birth years smoked at 19.9% at ages 15 to 19 in 2018 and at 30.4% at ages 20 to 24 in 2023[6]. Over the same decade the share smoking at ages 20 to 24 moved only from 34.0% to 30.4%, so growing up through the years of uptake, rather than change over time, accounts for most smoking among young adults. People aged 20 to 24 who had smoked had their first cigarette at a median age of 17, and 26.7% of ever-smokers aged 15 to 24 had theirs before the age of 15.

Fewer young people are starting, but fewer of those who start are stopping. Between 2018 and 2023 the share of people aged 20 to 24 who had never smoked rose from 63.1% to 68.3%, while former smokers fell from 3.5% to 1.3%; among those who had ever smoked, the share who had quit fell from 9.3% to 4.0%. The surveys do not record when a former smoker stopped, so a real decline in quitting cannot be told apart from a smaller and more committed group of young smokers.

Young smokers are almost all men: 58.4% of men aged 20 to 24 smoked in 2023, against 0.8% of women. The measure counts daily and occasional smokers, so it runs above the daily smoking of Table 13 (17.8% at ages 15 to 24 in 2023). Prevention has to reach secondary-school ages, where uptake happens; the national plan targets smoking at ages 10 to 21[4] and the 2024 regulation bans sales to people under 21[7].

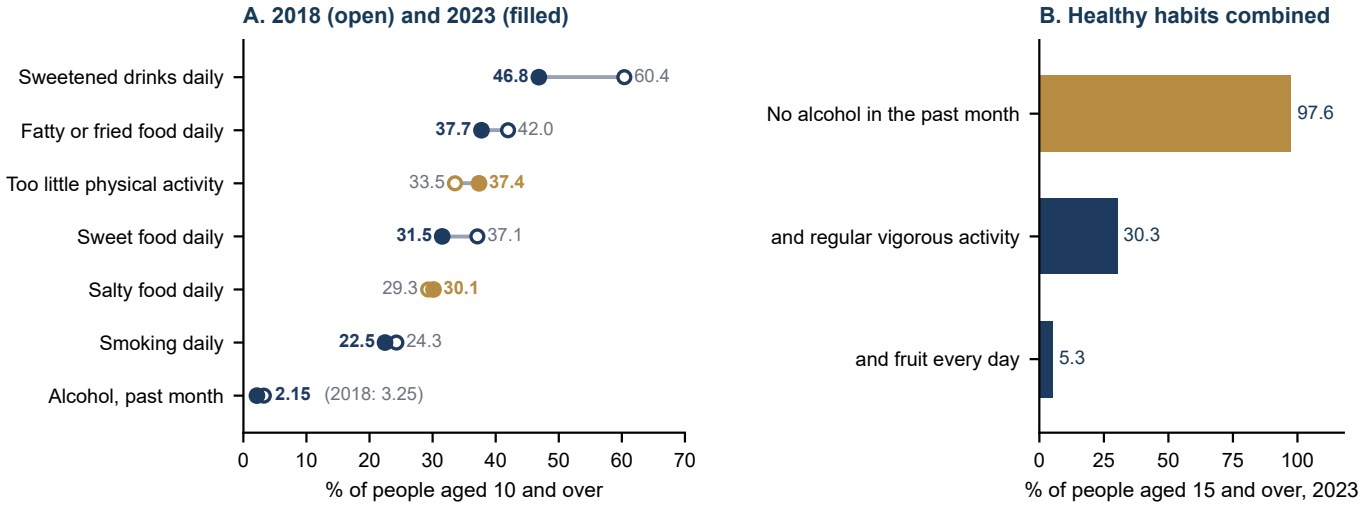
Sources: Ministry of Health, Riskesdas 2007, 2010, 2013 and 2018 and SKI 2023, ARC analysis of adolescents and young adults; RPJMN 2025–2029; Government Regulation 28/2024.

Risky Behaviours: Diet, Alcohol and Activity

Between 2018 and 2023 five of seven risky behaviours fell, while daily salty food and physical inactivity rose.

EXHIBIT 29

Five risky behaviours fell between 2018 and 2023 and two rose; few people combine healthy habits



A: people aged 10 and over; open dot 2018, filled dot 2023; navy: fell, gold: rose; every change shown has a 95% CI that excludes zero. Diet: eaten or drunk once a day or more, a count of how often, not of grams; 2018 to 2023 is the one interval in which all four diet questions were asked in the same words. 2018 diet and alcohol figures exclude records that lack a survey weight or design field. B: people aged 15 and over who answered for themselves; each bar adds one habit to the one above; regular vigorous activity is a different rule from the activity measure in A.

Table 15. Seven risky behaviours: what each survey question counts and which surveys it joins

Behaviour	The survey counts	Surveys read as one series	2023, % aged 10+
Sweetened drinks	drunk daily or more often	2018, 2023	46.8
Sweet food	eaten daily or more often	2018, 2023 (2007, 2013: one question with drinks)	31.5
Salty food	eaten daily or more often	2007, 2013, 2018, 2023	30.1
Fatty or fried food	eaten daily or more often	2013, 2018, 2023	37.7
Alcohol	any drink in the past month	2007, 2018, 2023	2.15
Physical inactivity	below the survey's activity rule	2013, 2018, 2023	37.4
Smoking	smoking every day	2013, 2018, 2023	22.5

A series joins only surveys whose questions name the same thing. Riskesdas 2010 asked none of the diet questions. No diet question measures grams of sugar, salt or fat. Exhibits 30 to 34 take one behaviour each.

Improvements in health behaviours have been uneven. Between 2018 and 2023, five of seven risky behaviours declined, but daily salty-food consumption and physical inactivity increased. Daily consumption of sweetened drinks fell from 60.4% to 46.8%, fatty or fried food from 42.0% to 37.7%, and sweet food from 37.1% to 31.5% among people aged 10 and over. Daily smoking declined from 24.3% to 22.5%, and past-month alcohol use from 3.25% to 2.15%. In contrast, daily salty-food consumption rose from 29.3% to 30.1%, and physical inactivity from 33.5% to 37.4%[8-10].

Risky behaviours also cluster. In 2023, 58.6% of people aged 3 and over consumed sweet, salty, or fatty food daily, and 11.9% consumed all three[8]. Although 97.6% of adults aged 15 and over reported no recent alcohol use, only 30.3% also engaged in regular vigorous activity, and just 5.3% also ate fruit daily (AMHASS). Fruit and vegetable intake averaged 1.8 portions per day; 96.5% consumed fewer than the five daily portions recommended by WHO. Dietary risks accounted for an estimated 13.6% of healthy life-years lost in 2016[11]. The surveys measure dietary frequency, not quantities consumed, and trends are compared only where question wording is consistent (Table 15).

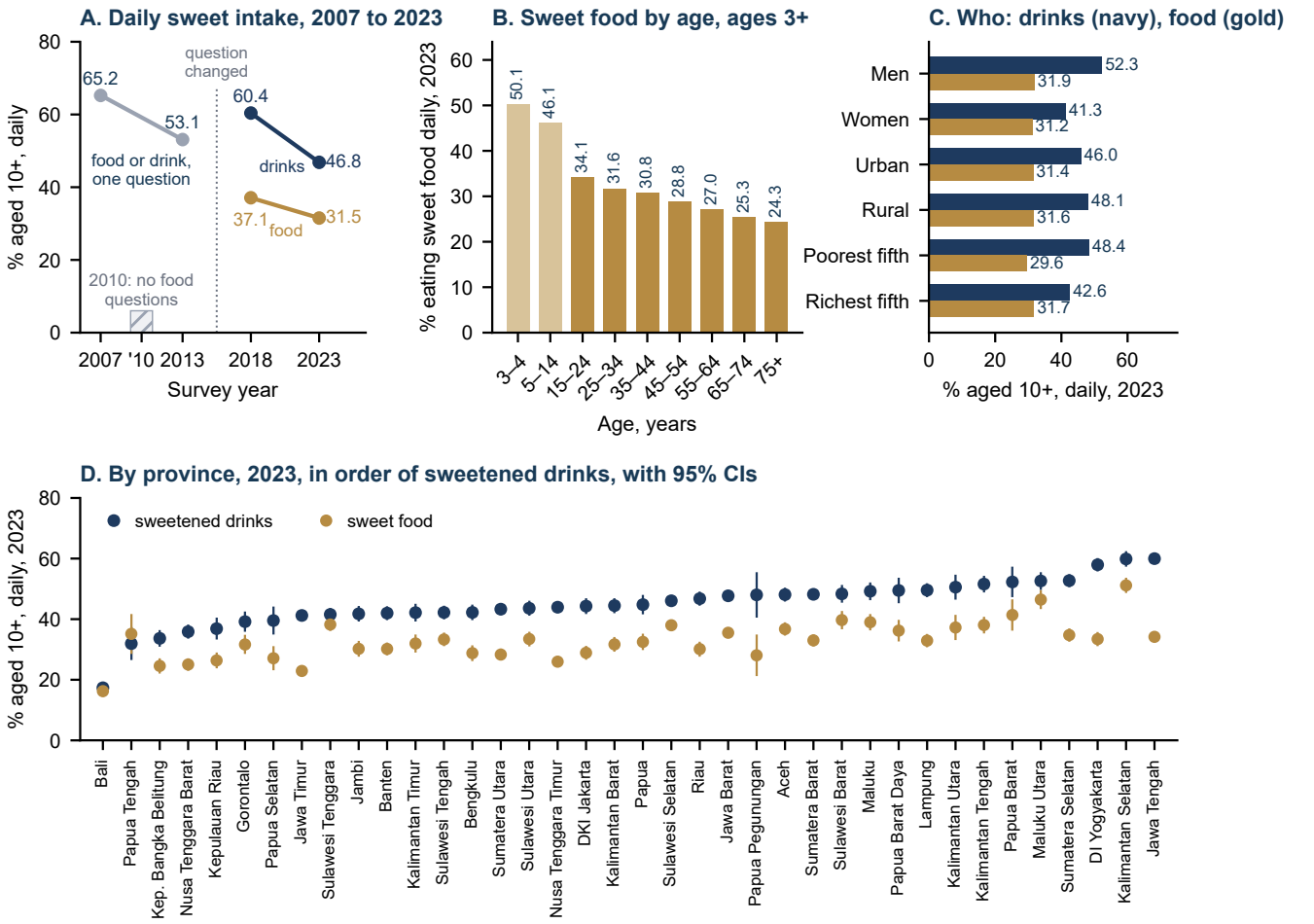
Sources: Ministry of Health, Riskesdas 2018 and SKI 2023, ARC analyses of diet, alcohol, physical activity and smoking; AMHASS; Global Burden of Disease 2016.

Risk Factor 3: Sweet Food and Sweet Drinks

Sweetened drinks are the most common daily habit measured, yet daily drinking fell from 60.4% to 46.8% after 2018.

EXHIBIT 30

Daily sweet drinks and sweet food fell after 2018; children eat sweet food most and men drink most



A: people aged 10 and over, 95% CIs; the surveys asked one question on sweet food or drink in 2007 and 2013 and separate questions on sweet food and on sweet drinks from 2018, so the lines before and after the change are not connected; Riskesdas 2010 asked no food questions. B: everyone the 2023 survey asked, from age 3; lighter bars: answers for children given by a caregiver. C and D: people aged 10 and over; wealth fifths of a household asset index. All: once a day or more, a count of how often, not of grams of sugar.

The surveys support three separate time series for sweet consumption: one combined measure of sweet foods and drinks in 2007 and 2013, and separate measures from 2018 onward. Daily consumption of sweet food or drinks fell from 65.2% to 53.1% of people aged 10 and over between 2007 and 2013. From 2018 to 2023, daily sweetened-drink consumption declined from 60.4% to 46.8% and daily sweet-food consumption from 37.1% to 31.5%, with declines across every age group. Sweetened-drink consumption also fell in 36 of 38 provinces[8].

Despite these declines, sweetened drinks remain common. In 2023, 47.5% of people aged 3 and over consumed them daily, including 19.8% more than once a day. Among those aged 10 and over, daily consumption was higher among men than women (52.3% versus 41.3%), among the poorest than the richest fifth (48.4% versus 42.6%), and among rural than urban residents (48.1% versus 46.0%). Each additional daily serving is associated with a higher risk of type 2 diabetes[12].

Sweet-food consumption is concentrated among children: 50.1% of children aged 3–4 consumed sweet food daily, compared with 24.3% of adults aged 75 and over. Estimates for young children rely on caregiver reports. Differences by sex and urban–rural residence were each less than one percentage point. Provincial variation was much wider: daily sweet-food consumption ranged from 16.2% in Bali to 51.1% in Kalimantan Selatan, while sweetened-drink consumption ranged from 17.4% in Bali to 60.0% in Jawa Tengah.

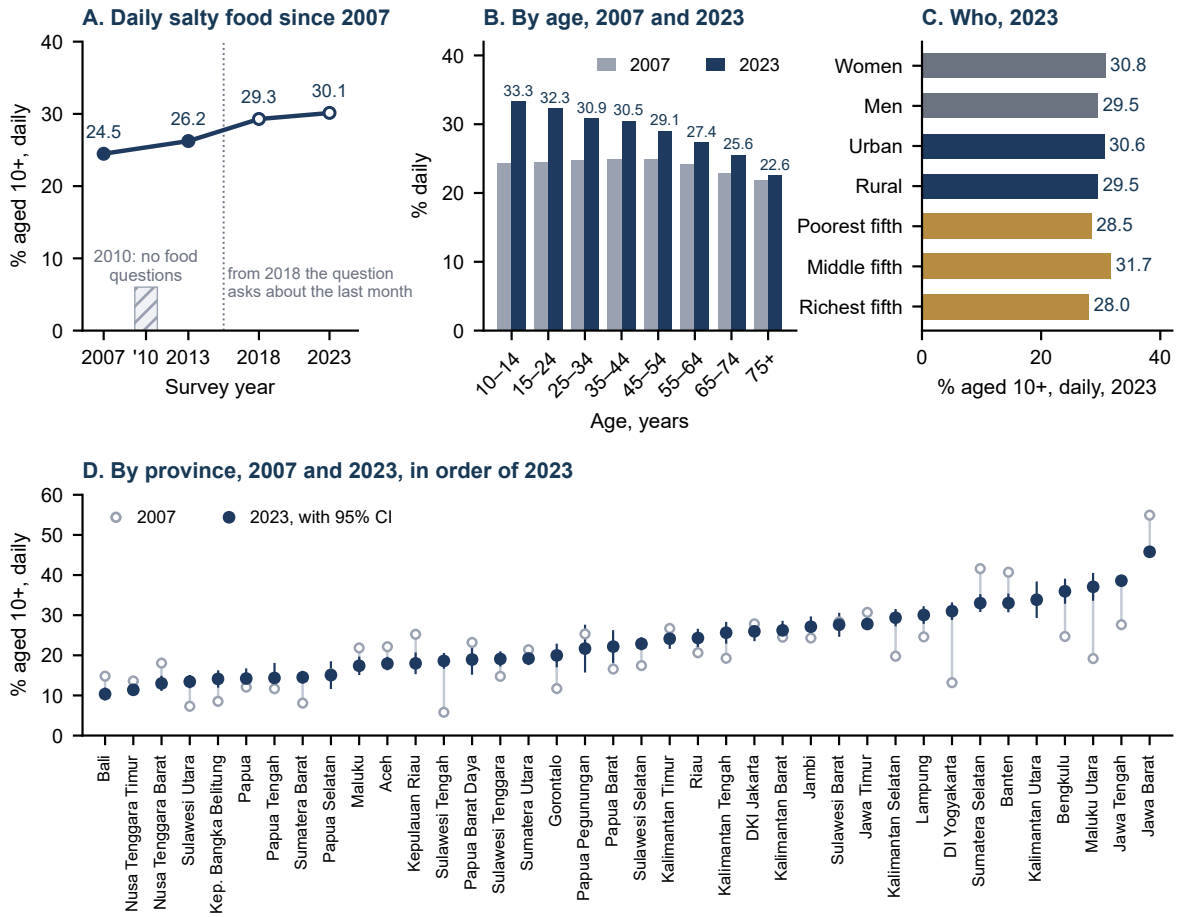
Sources: Ministry of Health, Riskesdas 2007, 2013 and 2018 and SKI 2023, ARC analysis of dietary risk behaviours; studies cited.

Risk Factor 4: Salty Food Since 2007

Daily salty food is the only diet item asked the same way since 2007, and it rose at every survey, to 30.1%.

EXHIBIT 31

Daily salty food rose from 24.5% to 30.1% between 2007 and 2023, most among the young



A: people aged 10 and over, 95% CIs; the question names salty food in the same words in all four surveys; hollow dots: from 2018 the question asks about the last month; Riskesdas 2010 asked no food questions. B: 2007 and 2023 by age. C: 2023; wealth fifths of a household asset index. D: Kalimantan Utara has no 2007 value. Once a day or more: a count of how often, not of sodium.

Salty-food consumption (*makanan asin*) is the only dietary behaviour tracked consistently across all four surveys since 2007. Among people aged 10 and over, daily consumption rose from 24.5% in 2007 to 26.2% in 2013, 29.3% in 2018 and 30.1% in 2023, an increase of 5.7 percentage points (95% CI, 5.2–6.2)[8]. The largest increase occurred between 2013 and 2018, when the question introduced a reference to the preceding month, but consumption also rose in the two intervals without changes in wording.

The increase was concentrated among younger people. Daily consumption among those aged 10–14 rose from 24.4% in 2007 to 33.3% in 2023, compared with a smaller increase from 21.9% to 22.6% among adults aged 75 and over. In 2023, women reported slightly higher consumption than men (30.8% versus 29.5%), while prevalence was highest in the middle wealth quintile (31.7%) and lowest in the richest (28.0%). Among people aged 3 and over, 12.8% consumed salty food more than once daily. Provincial trends varied: between 2007 and 2023, prevalence increased in 16 provinces, declined in 10 and showed no statistically significant change in 11.

These figures measure consumption frequency, not sodium intake, and ARC has no direct estimate of population salt consumption. Because reducing sodium lowers blood pressure and stroke risk[13,14], setting and monitoring salt-reduction targets requires direct intake measures, such as urinary sodium, alongside the limits established in the 2024 regulation (Exhibit 37).

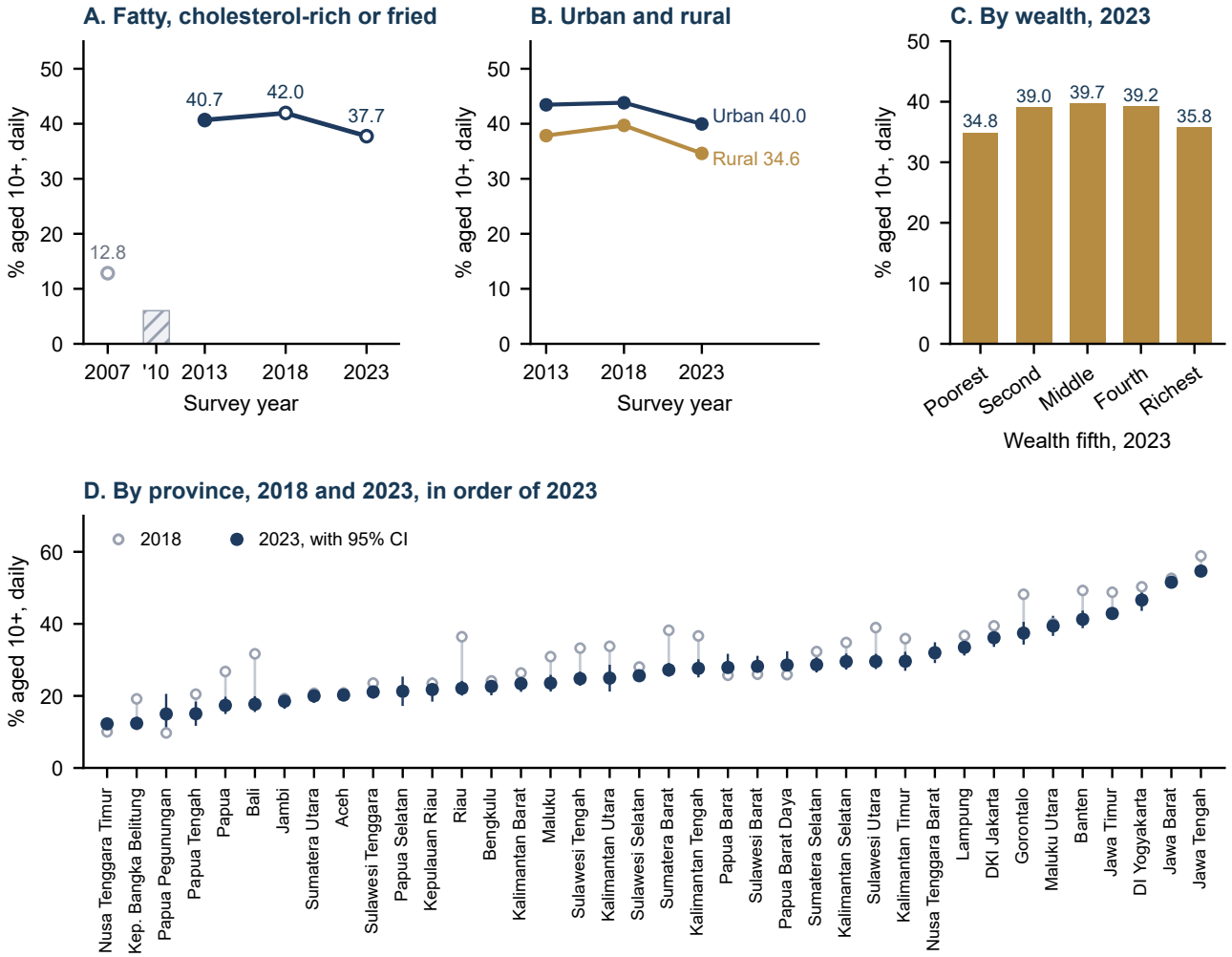
Sources: Ministry of Health, Riskesdas 2007, 2013 and 2018 and SKI 2023, ARC analysis of dietary risk behaviours; studies cited.

Risk Factor 5: Fatty and Fried Food

Daily fatty or fried food fell after 2018 but remains a habit of 37.7% of people aged 10 and over.

EXHIBIT 32

Daily fatty or fried food fell from 42.0% to 37.7% after 2018; towns and the middle fifths eat it most



A: people aged 10 and over, 95% CIs; the question names fatty, cholesterol-rich or fried food from 2013 (hollow dots: from 2018 it asks about the last month); the 2007 question named fatty food only and is not joined. B: 2013 to 2023. C: 2023; wealth fifths of a household asset index. Once a day or more: how often, not grams or type of fat.

The fatty-food series begins in 2013, when the survey question first included fried and high-cholesterol foods. Among people aged 10 and over, daily consumption rose from 40.7% in 2013 to 42.0% in 2018, before falling to 37.7% in 2023, a decline of 4.2 percentage points since 2018 (95% CI, 3.7–4.7)[8]. The 2007 question covered fatty food alone, making its estimate of 12.8% unsuitable for comparison. In 2023, 16.3% of people aged 3 and over consumed fatty or fried food more than once daily.

Daily consumption was consistently higher among urban than rural residents (40.0% versus 34.6% in 2023) and slightly higher among women than men (38.5% versus 37.0%). Prevalence was highest in the middle wealth quintile (39.7%), compared with 34.8% in the poorest and 35.8% in the richest. Prevalence also declined with age, from 39.8% among those aged 10–14 to 29.4% among adults aged 75 and over. Between 2018 and 2023, prevalence fell in 24 provinces and rose in one.

Consumption frequency did not reliably identify people with elevated blood lipids. In 2023, high LDL cholesterol was less prevalent among daily consumers than among those eating fatty or fried food less than once a week (Exhibit 79). The survey records frequency, not quantity or type of fat consumed. Such data can inform dietary health promotion but cannot substitute for direct measurement of blood lipids.

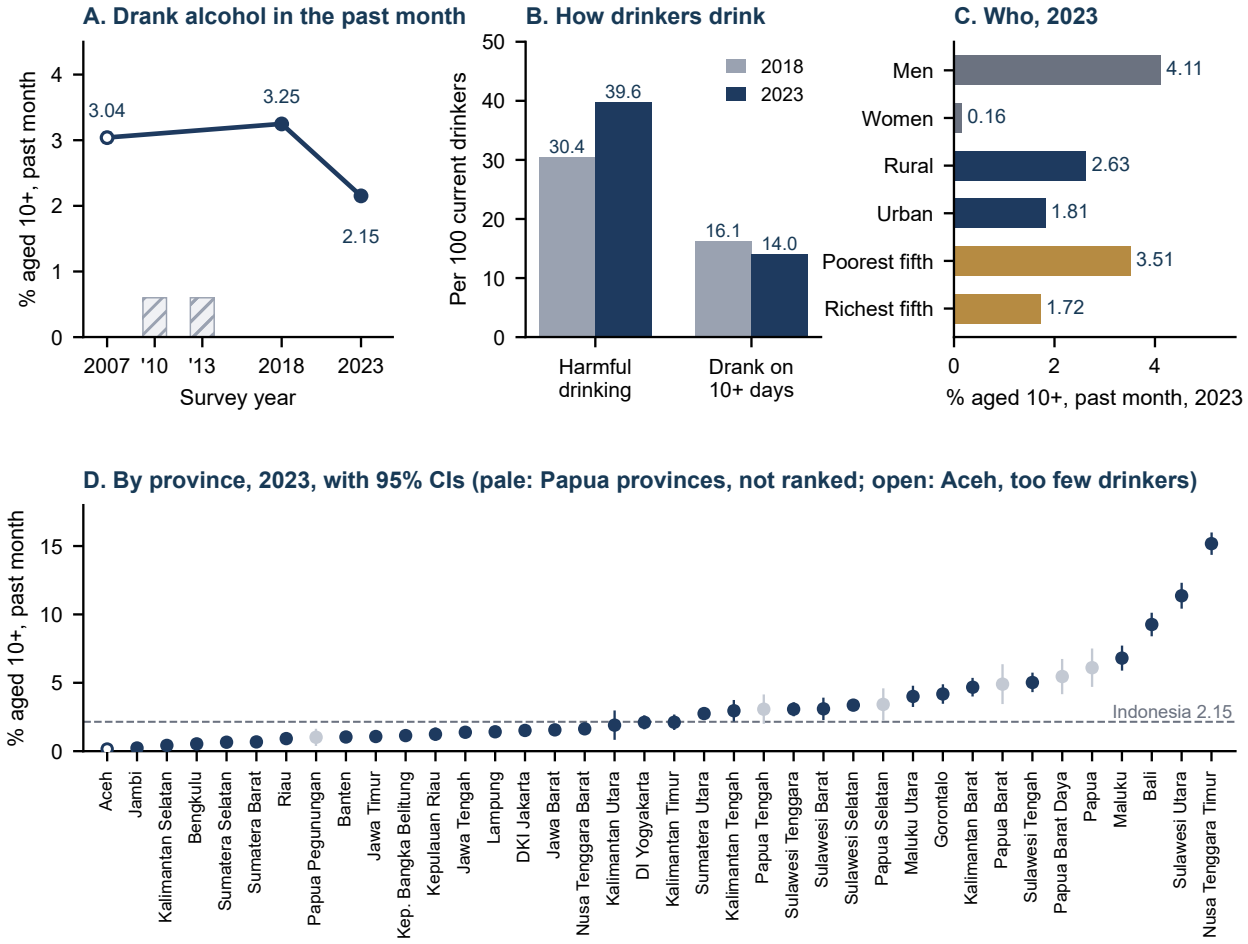
Sources: Ministry of Health, Riskesdas 2007, 2013 and 2018 and SKI 2023, ARC analysis of dietary risk behaviours.

Risk Factor 6: Alcohol

Fewer people drink alcohol than in 2018, but those who drink now drink more on each drinking day.

EXHIBIT 33

Fewer people drank alcohol in 2023 than in 2018, and harmful drinking among drinkers rose



A: people aged 10 and over, 95% CIs; hollow dot: routed through drinking in the past year; Riskesdas 2010 and 2013 carry no comparable item; 2018 leaves out 100,287 records with incomplete survey design fields (3.26% on all records). B: per 100 people who drank in the past month, from the drinks on a typical drinking day; no survey intervals. C: wealth fifths of a household asset index. D: Papua provinces printed, not ranked (boundaries unverified); Aceh rests on 28 drinkers.

Past-month alcohol consumption among people aged 10 and over was 3.04% in 2007, 3.25% in 2018 and 2.15% in 2023[9]. The decline after 2018 was 1.10 percentage points (95% CI, 1.00–1.19) and occurred across every age group and all 29 provinces with comparable estimates; the decline was statistically significant in 26 provinces. The 2007 estimate relied on a question routed through past-year drinking, making the small increase to 2018 less directly comparable than the subsequent decline.

Among those who continue to drink, consumption per occasion appears to have increased. Between 2018 and 2023, harmful drinking among current drinkers rose from 30.4 to 39.6 per 100, while the average number of standard drinks per drinking day increased from 6.1 to 12.4. Meanwhile, the proportion drinking on 10 or more days per month declined from 16.1 to 14.0 per 100.

These quantity estimates lack survey confidence intervals, and the calculated averages exceed published estimates; their absolute levels should therefore be interpreted cautiously, although the same conversion rule was applied in both years. Reducing alcohol intake can lower blood pressure among people who drink heavily[15].

Alcohol consumption was concentrated among men (4.11%, versus 0.16% among women), rural residents (2.63%, versus 1.81% among urban residents) and the poorest households (3.51%, versus 1.72% in the richest quintile). Across the 31 provinces with rankable estimates, prevalence ranged from 15.2% in Nusa Tenggara Timur to 0.23% in Jambi. The six provinces of Papua were excluded because their boundaries in the survey data could not be verified.

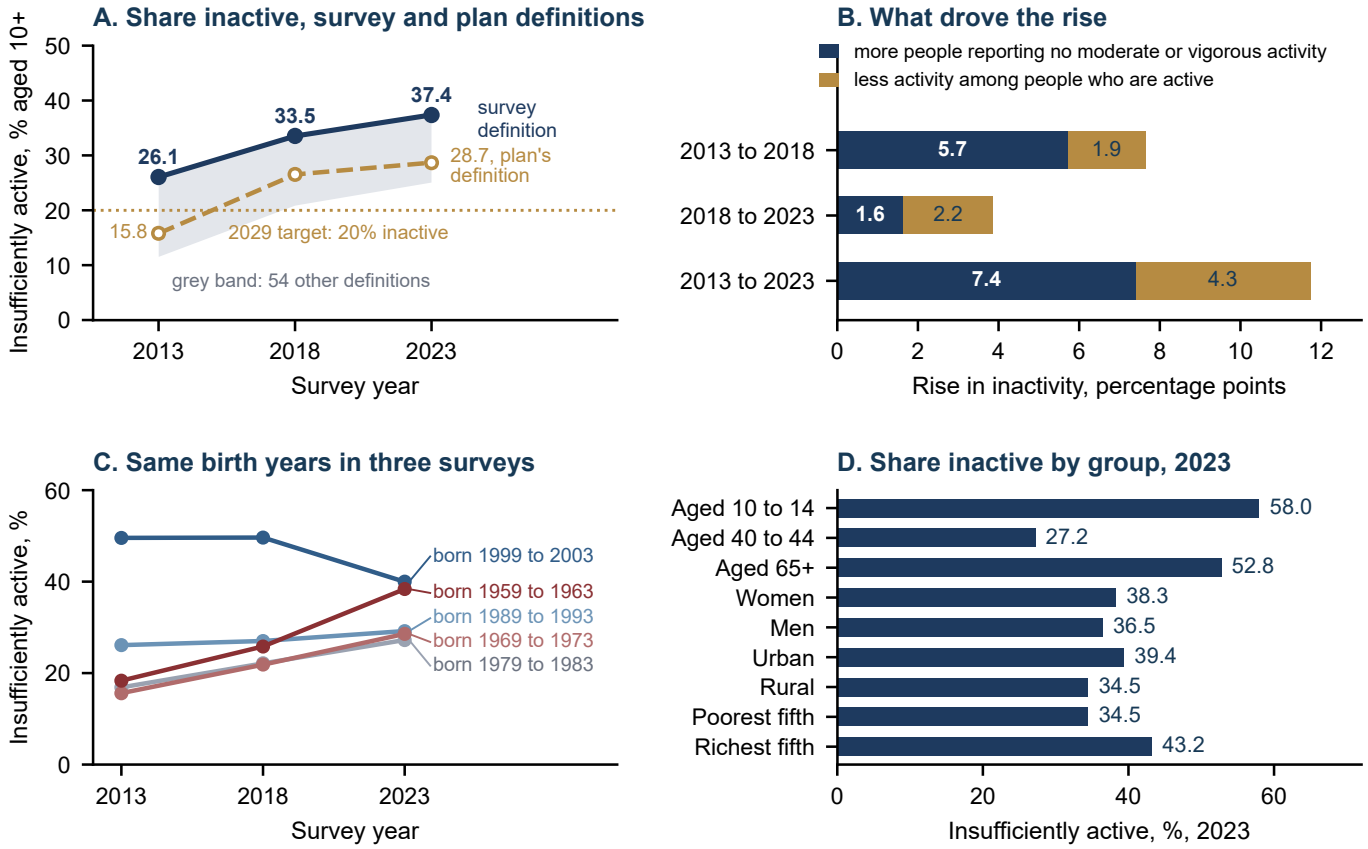
Sources: Ministry of Health, Riskesdas 2007 and 2018 and SKI 2023, ARC analysis of recent alcohol use; studies cited.

Risk Factor 7: Physical Inactivity

Inactivity rose from 26.1% to 37.4% between 2013 and 2023 under every definition, but its level depends on the rule.

EXHIBIT 34

Inactivity rose under every definition, mostly because more people reported no activity



Insufficiently active: people aged 10 and over. A: navy, the survey's definition (vigorous activity on 3 or more days and 1,500 MET-minutes a week, or moderate activity on 5 or more days and 150 minutes), with 95% CIs; grey band, the range under 54 other definitions applied to the same answers; gold, the plan's definition (150 minutes of activity a week in total) and the 20% inactive that its 80% target implies. B: the rise split into more people answering no to both opening questions (any vigorous activity, any moderate activity) and fewer minutes or days among those who answered yes. C: the same birth years in three separate surveys, not the same people, each ten years older in 2023 than in 2013 (born 1999 to 2003: aged 10 to 14 in 2013, 20 to 24 in 2023); deaths and migration not adjusted. D: wealth fifths of a household asset index (ARC harmonisation).

Insufficient physical activity among people aged 10 and over rose from 26.1% in 2013 to 33.5% in 2018 and 37.4% in 2023[10]. Although estimates vary substantially across alternative definitions, all six definitions used in practice show an increase of 11.3–14.1 percentage points, which cannot be explained by population ageing alone. Under the published definition, the increase reflects both more respondents reporting no activity in the initial screening questions and changes in reported activity among the remainder. The data cannot distinguish actual behavioural change from differences in question administration after Riskesdas 2018 was integrated with Susenas.

The definition matters for tracking national targets. The plan's baseline of 62.6% with adequate activity is the complement of the published 2023 inactivity estimate[16]. However, applying the plan's own threshold of 150 minutes per week, without a minimum number of active days, yields a lower inactivity estimate of 28.7% in 2023[10]. Progress towards the 2029 target of 80% with adequate activity should therefore be assessed using the plan's definition.

Insufficient activity was highest among adolescents aged 10–14 (58.0%) and adults aged 65 and over (52.8%), and more prevalent in urban areas than rural areas (39.4% versus 34.5%). Provincial estimates ranged from 27.8% in Jawa Timur to 62.0% in Papua Tengah, but rankings changed considerably between 2013 and 2023 (rank correlation 0.48). Provincial differences should therefore be interpreted cautiously, particularly when assessing changes over time.

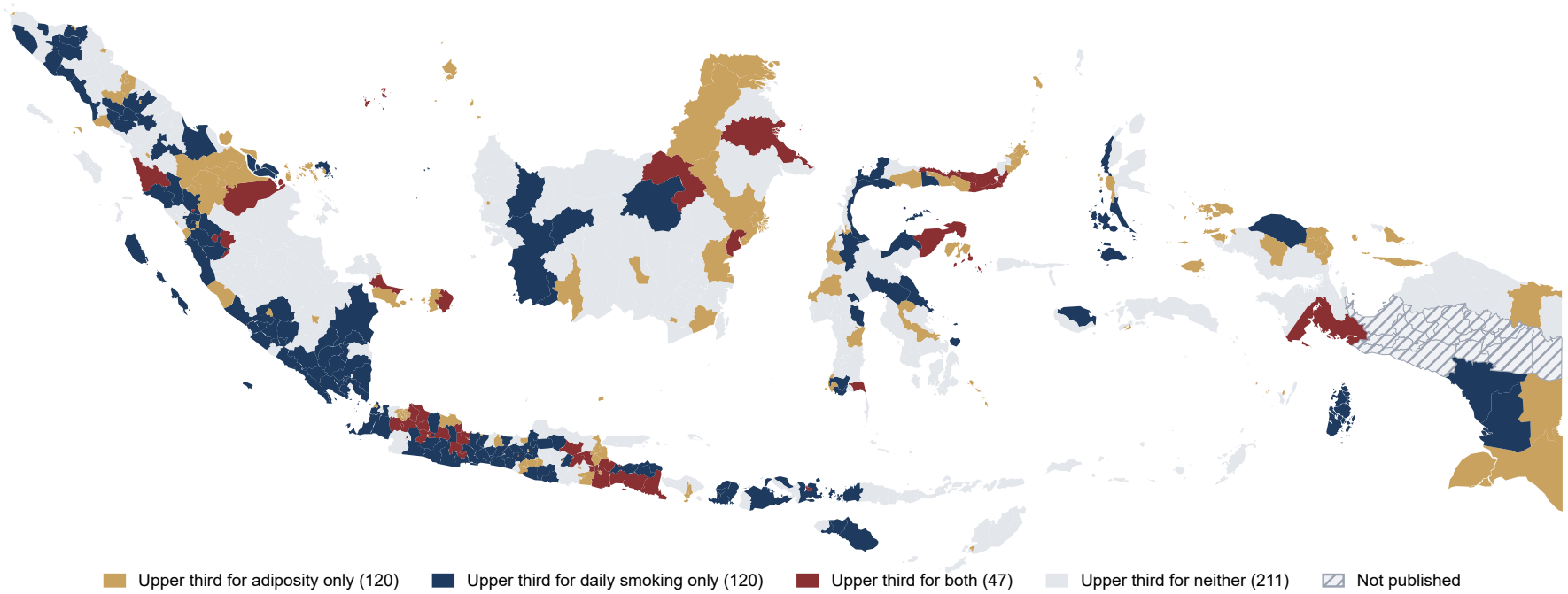
Sources: Ministry of Health, Riskesdas 2013 and 2018 and SKI 2023, ARC analysis of physical activity (wealth: ARC harmonisation); RPJMN 2025–2029, Lampiran II.

Excess Body Fat and Daily Smoking by District

In 2023, the two risks often fall in different districts.

EXHIBIT 35

In 2023, adiposity was concentrated in cities and daily smoking in regencies; only 47 of 498 districts were in the upper third for both



Adiposity: overweight (BMI 25 or more) with central obesity, adults 18+. Daily smoking: people aged 10+. Thirds of the districts with a published estimate; hatched: not published. Source: Ministry of Health, SKI 2023, AMHASS district estimates.

Table 16. Districts in the upper third of each risk, by type, 2023

Districts	Adiposity only	Daily smoking only	Both	Neither	All banded
Cities	65	2	11	20	98
Regencies	55	118	36	191	400
Cities, % of the column	54.2	1.7	23.4	9.5	19.7

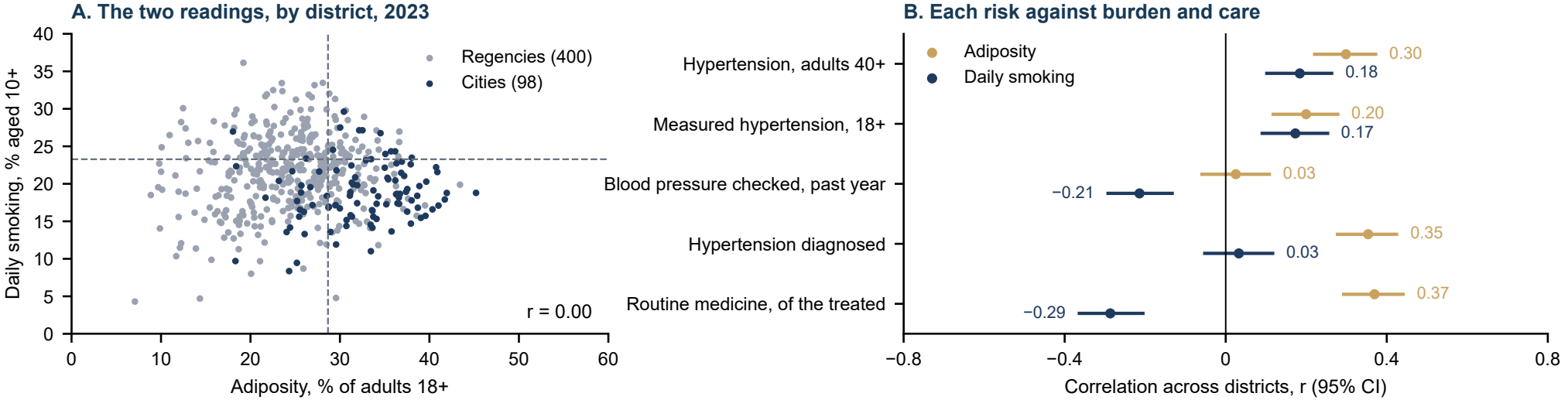
Upper third: the top third of the districts with a published estimate for each reading.

Excess Body Fat and Smoking in Cities and Regencies

Excess body fat is higher in cities and smoking is higher in regencies.

EXHIBIT 36

Across districts in 2023, adiposity went with more hypertension (r = 0.30) and daily smoking with fewer blood pressure checks (r = −0.21)



A: each dot is a district; dashed: lower limits of the upper thirds. B: Pearson r (a measure of how closely two things move together), 95% CI by Fisher's z; diagnosed: share of adults ever diagnosed with hypertension.

Daily smoking was 21.9% of people aged 10 and over in regencies and 18.5% in cities. Excess body fat is a city risk and daily smoking is a regency risk. A median 33.5% of adults are overweight and centrally obese in the 98 cities, compared with 24.5% in the 400 regencies. Daily smoking is 18.5% in cities and 21.9% in regencies (Mann-Whitney test, $p < 0.001$ each; Ministry of Health, SKI 2023). Across 498 districts the two are unrelated (r = 0.00).

Each risk goes with a different weakness in care. Within cities and within regencies, the two rise together only weakly (r = 0.13 and 0.13). Adiposity is moderately linked with hypertension at 40 and over (r = 0.30) and with more diagnosis (0.35). Districts with more daily smokers check blood pressure less (−0.21), and fewer of their treated patients take medicine routinely (−0.29).

These correlations describe places, not individual people, as in a 2018 district analysis[17]. Prevention therefore needs two maps. Cities face excess body fat and the raised blood pressure that comes with it.

Regencies face smoking with weaker screening and care; the median regency has 21.9% smoking daily (panel A). Quitting support therefore belongs with the free health check there. National measures reach both (Exhibit 37); only 47 districts need both at full strength.

Sources: Ministry of Health, SKI 2023, AMHASS and harmonised district estimates, ARC analysis.

Policy Tools and the National Plan

The national plan sets targets for smoking but not for food.

EXHIBIT 37

The national plan targets smoking (8.4% by 2029) but not sugar, salt or fat

Table 17. Policy levers on cardiometabolic risk: legal basis and national plan targets

Lever	Legal basis	What it does	National plan target	Exposure, 2023
Tobacco excise	RPJMN 2025–2029; Finance Minister Regulation 97/2024	Gradual excise rises, fewer tariff tiers; 2025 raised minimum prices, not tariffs	Smoking at 10-21: 12.4% to 8.4% (2029)	22.5% of people 10+ smoke daily; no survey value for ages 10–21
Smoke-free areas (KTR)	Law 17/2023; Government Regulation 28/2024, Art. 442	No smoking, sale or promotion of tobacco in designated areas, set by district regulation	KTR districts: 367 to 514 (2029)	147 districts to go
Tobacco sale and advertising	Government Regulation 28/2024, Art. 434, 438, 446[7]	No sale under 21 or of single sticks; warnings on 50% of packs; no social-media advertising	None	Not measured
Sugar, salt and fat in food	Government Regulation 28/2024, Art. 194-196; Health Minister Regulation 30/2013; Health Minister Decree HK.01.07/MENKES/301/2026[18]	Maximum content to be set; labels; marketing limits above it; low sugar, salt and fat zones run by districts; Nutri-Level grades A to D on sweetened ready-to-eat drinks (2026)	None	30.1% eat salty food daily
Excise on sweetened drinks	Government Regulation 28/2024, Art. 194(4)	No excise in force (2025); Art. 194(4) allows one	None	46.8% drink sweet drinks daily
Physical activity	RPJMN 2025–2029, risk-factor control	Health promotion and community activity	Adequate activity: 62.6% to 80% (2029)	37.4% inactive on the survey's rule; 28.7% on the plan's own rule

RPJMN: national medium-term development plan 2025–2029. Exposure: harmonised SKI 2023 estimates (Exhibits 27, 30, 31 and 34) or plan arithmetic.

Tobacco control measures are established in national law and planning, but implementation remains uneven. The national development plan (RPJMN) commits to gradual excise increases and targets reducing smoking among people aged 10–21 from 12.4% to 8.4%, alongside smoke-free regulations in all 514 districts, up from 367[16]. However, the 2025 policy adjustment raised minimum prices rather than excise tariffs[19]. International evidence shows that tax increases are particularly effective in reducing smoking among young people and lower-income groups[20]. Among 244,602 JKN members, cardiovascular mortality was lower in provinces with stronger smoke-free policies (relative risk 0.83)[21].

Food policy measures are more recent and less developed. A 2024 health regulation provides for central government limits on sugar, salt, and fat in processed and ready-to-eat foods, restrictions on marketing products exceeding those limits, and possible excise on selected processed foods[22]. Nutrition labelling has been required since 2013[23], while from April 2026, large ready-to-eat food businesses must grade sweetened drinks from A to D using Nutri-Level labels[18].

Yet these measures lack explicit national targets, despite daily sweet-drink consumption of 46.8% and daily salty-food consumption of 30.1% among people aged 10 and over. International experience offers relevant evidence: taxes on sweetened drinks reduced sales of taxed products by 15%[24], while England's salt-reduction programme lowered daily salt intake by 1.4 g[25]. These results illustrate potential policy effects, not forecasts for Indonesia.

Three priorities follow. First, the Ministry of Finance should resume annual excise tariff increases that outpace income growth, as envisaged in the national plan. Second, the Ministry of Health and the National Agency of Drug and Food Control (BPOM) should publish and implement the sugar, salt, and fat limits and front-of-pack labelling provisions. Third, provinces should extend smoke-free regulations to the 147 districts still without them before 2029. Physical activity also needs leadership beyond the health sector, and one definition: the plan's 62.6% baseline comes from the survey's rule, while the plan's own indicator would count 28.7% as inactive in 2023, not 37.4% (Exhibit 34); progress towards the target is judged on the plan's own rule.

Sources: RPJMN 2025–2029; Government Regulation 28/2024; harmonised SKI 2023, ARC analysis; studies cited.

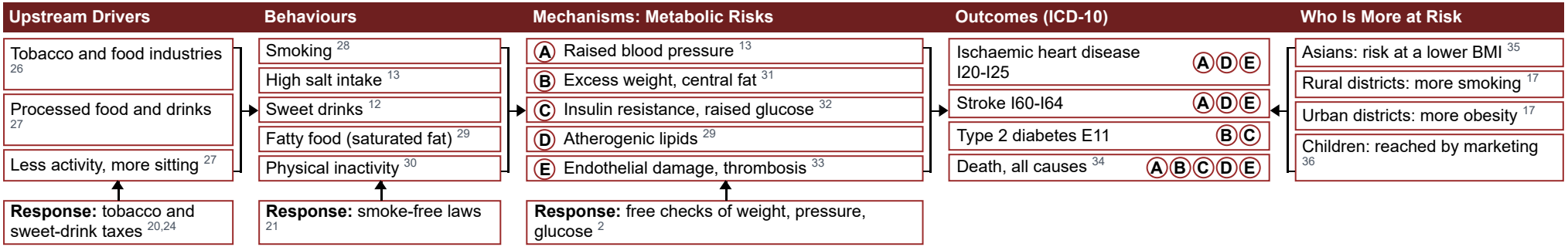
Framework of Risk Factors and Prevention

Research links smoking, salt, sweet drinks, saturated fat, inactivity and excess weight to heart disease, stroke and diabetes.

The links run through blood pressure, blood sugar (glucose), blood lipids and damage to the arteries. Taxes and smoke-free laws act earlier in the chain, on the causes; the strategies and targets behind them open the chapter (Exhibit 127).

EXHIBIT 38

Pathways from behaviours and weight to heart disease, stroke and diabetes, from the literature



Arrows show possible pathways taken from the cited reviews. They are not effects estimated by ARC. Letters link each outcome to its causes. Effects from other countries are theirs, not estimates for Indonesia. Strength of evidence: established = a WHO guideline or technical document, a pooled effect with I-squared (a measure of how much studies disagree) under 75%, or a burden model for its own burden; probable = more disagreement between studies or none reported, a review without a pooled estimate, or several consistent studies; emerging = few or conflicting studies; single = one study or model; none = no review or study found.

Table 18. Behavioural and weight risks: strength of the evidence and Indonesian studies

Risk: outcome (ICD-10)	Pathway	Strength	Main review: headline effect (95% CI)	Indonesian studies found
Smoking: coronary heart disease (I20-I25)	D E	Probable	HR 1.60 (1.49 to 1.72) in smokers[28]	3: OR 2.03 in former smokers[37]
High salt: stroke (I60-I64)	A	Probable	Higher sodium: RR 1.24 (1.08 to 1.43)[13]	None on stroke; 6 on blood pressure, 5 positive[38]
Sweet drinks: type 2 diabetes (E11)	B C	Probable	RR 1.18 (1.09 to 1.28) per daily serving[12]	3: more drink calories, more diabetes[39]
Saturated fat: cardiovascular events	D	Established	Less saturated fat: RR 0.79 (0.66 to 0.93)[29]	2, mixed: fried food, dyslipidaemia RR 1.35[40]
Inactivity: type 2 diabetes (E11)	B C	Established	High activity: RR 0.65 (0.59 to 0.71)[30]	4: active, aOR 0.71 (0.61 to 0.82)[41]
Obesity: type 2 diabetes (E11)	B C	Probable	Versus normal weight: RR 4.56 (3.69 to 5.64)[42]	2 cohorts: rising BMI predicted diabetes[43]

Main review: pooled cohorts unless stated; RR or HR above 1 means more events with the risk, below 1 fewer with the healthier habit. Indonesian studies: PubMed search per risk, 18 Oct 2026, abstracts read; blood pressure counted for salt, glucose for sweet drinks, lipids for fat.

Sources: reviews and Indonesian studies as cited; PubMed search, 18 Oct 2026.

About This Chapter

Risk Factors and Prevention Policy, chapter 3 of Indonesia Cardiometabolic Disease: An Epidemiologic Diagnostic of Burden, Risk Factors, Primary Care and Referral Care Management.

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